How to Think Different™

Using macOS as a developer: a mini-workshop



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Table of Contents

I. Introduction
II. OS navigation
III. Keyboard shortcuts
IV. Pitfalls and troubleshooting
V. Third-party tools and apps
VI. Third-party extensions
VII. Useful built-in applications
VIII. Saner settings
IX. Dotfiles

I. Introduction

What this workshop offers

I am kind of a perfectionist and like to customize everything.

What I will share with you is the result of 10 years of perfecting my setup.

The focus will be:

- A highly opinionated way of using macOS,
- Removing barriers to productivity,
- Having a neat, customized OS and tools.

You don't have to take everything

I'm just showing you my way of working:

- Keep what you like
- Knowing about the rest will still help

Practice makes perfect

This workshop will:

- What to avoid.
- Teach a lot of keyboard shortcuts.

It will take some time to stick, come back to these slides.

(And don't hesitate to contact me to ask questions!)

My history with computers

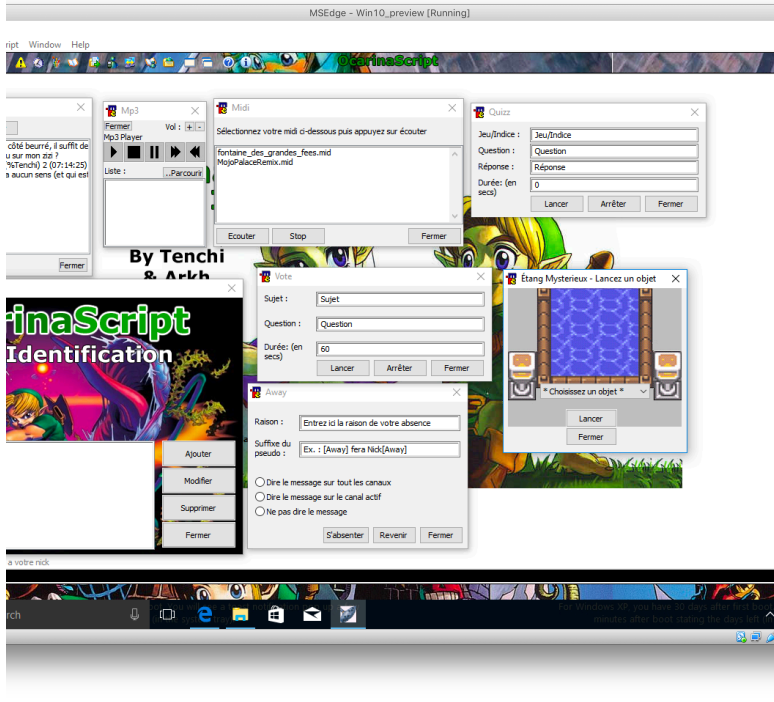


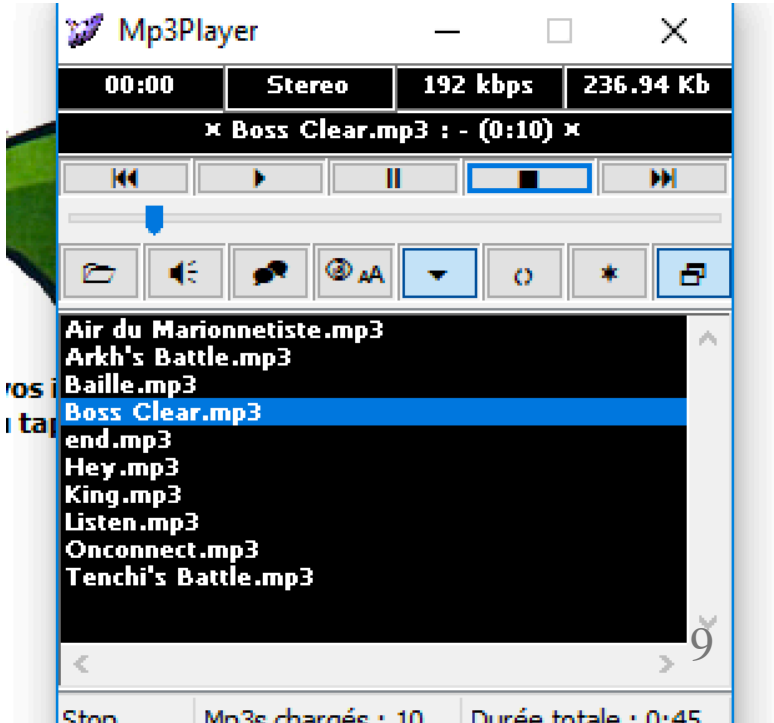
Phase 1

- Started using a Mac at age 3-4
- Macintosh SE/30
- 1 MB of RAM (extended to 2 MB)
- 512 × 342 black and white
- Classic Mac OS:
  not suited at all for developers

Phase 2

- Switched to windows at around age 10
- Age 11: made my own mIRC script client with ~100 users
- First Python project at age 12: Pong clone
- Random explorations of Linux many times
  - Windows is OK for dev. Experience is unstable
  - Linux is great for dev but needs a lot of care and troubleshooting







Phase 3

- 2015: Started at Ordina's *“Scala Team”*
- Only team at Ordina with Macbooks
- Mac OS X was based on Free BSD
- With some tweaks and getting used to:
  - UNIX-like experience and easy to use
  - Great for dev! The ups of Windows and Linux without their downs.

II. OS navigation

It's not what it looks like! I can explain!

MacOS Usage overview

- Two main modes of usage:
  i. Casual: full screen apps, multiple spaces, gestures.
  ii. Power user: all in one screen, keyboard shortcuts (almost no gestures).
- Many features aimed at casual use can make things *frustrating*:
  This gives macOS a casual reputation.

→ We're going to curate features and learn some power tools.

Basic window management

- Traffic light buttons: 
  - **Red**: close,

    Note: Closing the last window of an app most often doesn't quit the app.
  - **Yellow**: minimize,

    Recommendation: never use, forget about minimizing.
  - **Green**: full screen.

    Recommendation: use rarely, remember that it makes a new space.
- Double click title bar: maximize (we'll come back on this later).
- Exposé: overview of *all windows* or *current application*.

A note about closing windows

- On Windows, closing an app's main window exits that app.
- On macOS, the app keeps running:
  - Focusing that app shows no window, but the menu bar has its name and menu items.
  - This results in a lot of zombie apps that take ram and do nothing.
  - Only by quitting the app with ⌘ + Q or using the menu actually quits the app.

Don't close windows, just use ⌘ + Q.

A note about minimization

- On Windows, minimizing a window just hides it.
- On macOS, minimizing a window puts it in a special prison cell:
  - It disappears from alt-tab,
  - It disappears from Exposé,
  - It literally goes to prison, in the tiny tiny dock.

Don't minimize windows, they will be sad, and you too.

Spaces

- Each space has its own windows.
  - Alt tabbing between apps that are in multiple spaces can be confusing.
- Each full screen *window* is in its own space (even a full screen YouTube video).
- Exposé part 2: move windows to other spaces.

Recommendation: never use spaces, unless you need one app in full screen (for a presentation).

A note about full screen

- On Windows, clicking the “maximize” icon (here the green ball ) just makes the window use as much space as possible on the screen.
- On macOS, it puts the application in full screen. Full screen apps go to their own dedicated space:
  - It disappears from alt-tab,
  - It disappears from that app's Exposé (but goes to the top of the Exposé screen).

Don't full screen windows, unless you want to look neater during a presentation.

A note about the traffic light buttons 

Recommendation: avoid using.

Spotlight

One of the most useful shortcuts: `⌘` + `Space`.

Use it for everything:

- Launching apps (launching same app switches focus to it),
- Searching files,
- Quick web search.

Recommendation: use it all the time.



The desktop

Recommendation: never use.

The dock

- Basically just a list of app shortcuts.
- Not like the Windows taskbar, each icon is one application, not window.
- “Minimized” windows go down there on the right: this is not useful at all, avoid minimizing.

Recommendations:

- Can be used for quick app shortcuts, but just use Spotlight instead.
- Keep it hidden for maximal screen real estate.

Finder

The “Explorer” of macOS. It's better because you don't explore, you find.

Worthy of note:

- You can't exit that the Finder. It runs forever and `⌘` + `Q` does nothing.
- Supports tabs (`⌘` + `T`).
- Dragging files moves, hold `⌥` to copy.
- Use `Space` to preview things (more on that on the *Preview* slide).

III. Keyboard shortcuts

A frankly unreasonable amount of shortcuts.

Why keyboard shortcuts

Our focus today:

- Navigating the OS without clicking around,
- Editing text efficiently,
- Being in control and overall productivity.

It will feel like a lot, but I didn't look up any of these: it's really the list of stuff I use daily.

Note: The following slides apply OS-wide.

The cast of characters

In order of awesomeness (subjective).



Command

Symbolized as ⌘

Most main actions (or *commands* see what they did there) of an app use ⌘ (and not the *Control* key like in Windows).

Fun facts:

- Chosen by Susan Kare, designer of the icons in classic Mac OS and the Solitaire cards in Windows 3.1.
- In Scandinavia: denotes a place of cultural interest.

Control

Symbolized as ^

Macs have both *Command* and *Control*,
and it makes them keyboard-shortcut monsters.

- Windows and Linux barely use their *Meta* key, and its usage is for global shortcuts.
- On Mac, *Command* is completely application-oriented, tripling keyboard shortcuts:
  - ⌘ + …
  - ^ + …
  - ^ + ⌘ + …

Fun fact: the ^ is used on road signs to indicate roofed infrastructure.





Option

Symbolized as ⌥

Option is the macOS name for *Alt*:

- Used less for keyboard combinations (but still used).
- ⌥ + any letter makes symbols.
- Rarely used for shortcut combos.

Fun fact: I'm *really* struggling to keep the road sign theme going on.

Fn

Symbolized as `Fn`

- Simulate missing keys (Page up, home, F-keys, etc.).
- Switch keyboard layouts.

Note:

- On Windows laptops, `Fn` is just a hardware-level key to compensate for missing keys.
- On macOS, `Fn` is actually handled at the OS-level and is used in some keyboard shortcuts.



Yep I give up, best I could find.

Fun fact: *It's a weapons factory in Belgium.*

The shortcuts

In order of whatever (subjective).

Keyboard window management

Shortcut	Action
⌘ + Tab	Tabbing through **applications** (⇧ : reverse).
⌘ + < / >	Tabbing through **windows**.
⌘ + W	Close window.
⌘ + ⇧ + W	Close all windows in current app.
⌘ + Q	Quit application. Recommended: disable in some important apps.
^ + ⌘ + F	Toggle full screen (same as green button).

Clipboard

- Keyboard: `⌘` + `X` / `C` / `V`: Cut / Copy / Paste.

  Note: since copying is bound to ⌘, this means that `^` + `C` is still available!

- Terminal: `pbcopy` and `pbpaste` commands, very useful:

```
cat slide.md | pbcopy  # copies the contents of slide.md into clipboard
pbpaste > new_slides.md  # pastes the contents of the clipboard into new file
echo "Hello, $(pbpaste), bye bye" > wrapped_slides.md  # Can be used like any command
```

Classic UNIX keyboard shortcuts

MacOS inherits some UNIX terminal keyboard shortcuts **OS-wide**.

That's right, not just in terminals, but **everywhere you can type text**.

Shortcut	Action
^ + A / E	Jump to beginning/end of line.
^ + T	Transpose letters.
^ + B / F	Move one character left/right.

Moving around text

Shortcut	Action
Fn + ↑ / ↓	Page up/down.
Fn + ← / →	Home/end.
Fn + ⌫	Erase to the right (like Del).
⌥ + ← / →	Move cursor a whole word left/right. Combine with ⇧ for selection.
⌘ + ← / →	Move to beginning/end of physical. line (behavior slightly different than ^ + A / E).

Searching

Shortcut	Action
⌘ + F	Start search.
⌘ + G	Next result.
⌘ + ⇧ + G	Previous result.

Note: Shift ⇧ often inverses the effect of a shortcut (or expands its scope, like "Close all windows").

Screenshots

Prepare for some carpal tunnel syndrome:

Shortcut	Action
⌘ + ⇧ + 3	Capture whole screen, save to configured folder.
^ + ⌘ + ⇧ + 3	Capture whole screen, save to clipboard.
⌘ + ⇧ + 4	Capture selection (drag mouse), save to configured folder.
^ + ⌘ + ⇧ + 4	Capture selection (drag mouse), save to clipboard.
⌘ + ⇧ + 5	Screen recording tool.
^ + ⌘ + Esc	Stop recording.

Capture selection to clipboard

You're probably thinking: *"WTF."*

1. First place your thumb on ⌘,
2. Then place your pinky on ⇧,
3. Then place your ring finger on ^,
4. Finally, flatten your hand and reach for 4 with your index finger.
5. You can now release the keys!
6. Drag a box with your mouse and release.

Pro tip: while dragging the box, keep holding mouse click and hold Space to reposition the box.

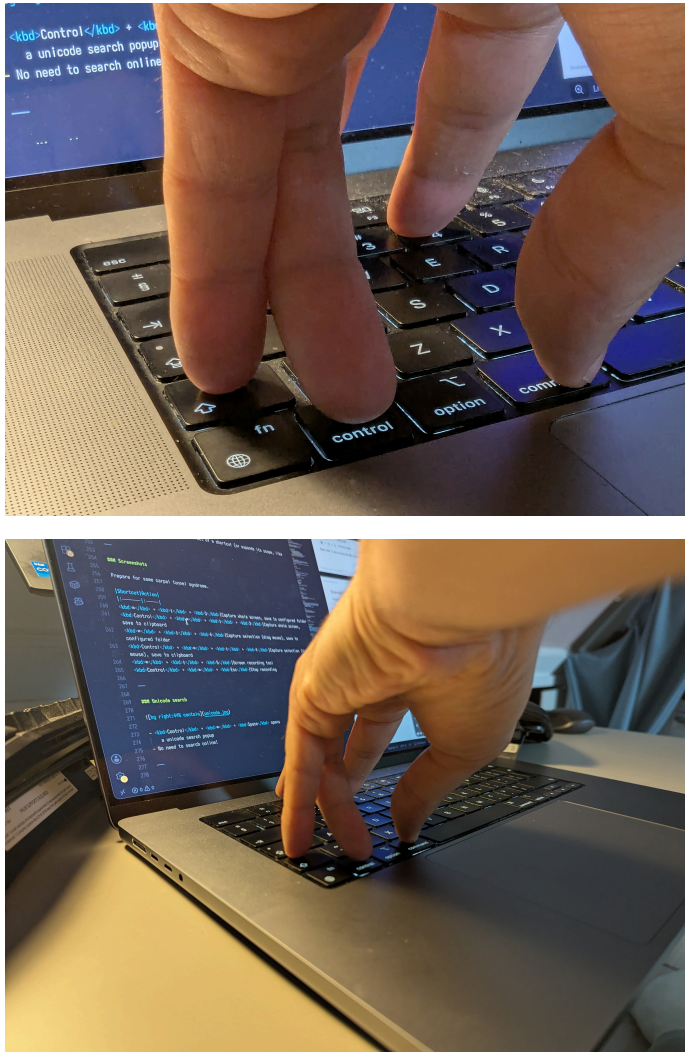


Figure 38.a and 38.b: How to RSI



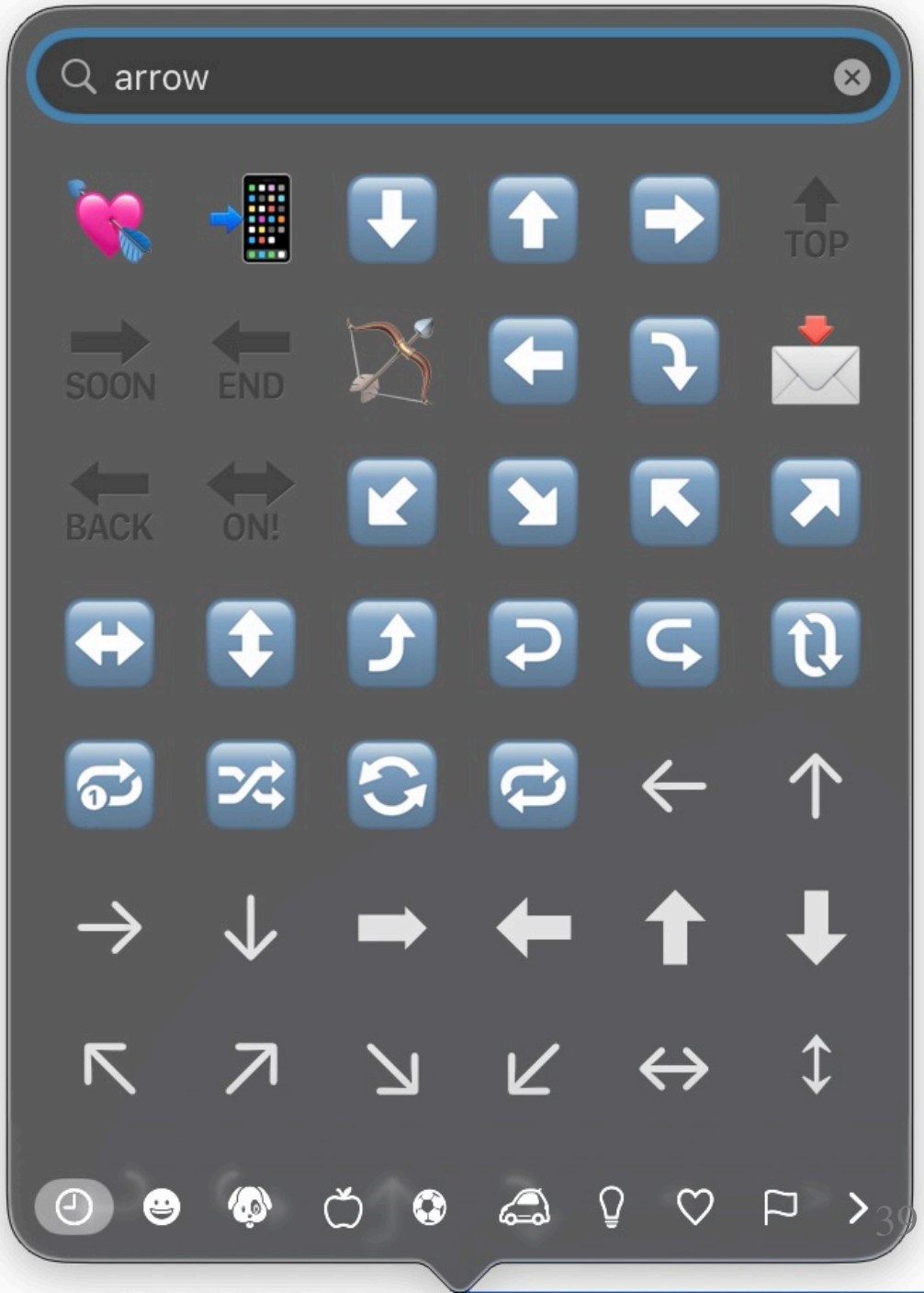
Window mode

In mode 4 (drag a box thingy), press `Space` to switch to window capture mode.

- Captures a single window,
- With PNG transparency and shadows,
- Isolated from anything in the background.

Unicode search

- ^ + ⌘ + Space opens a Unicode search popup.
- No need to search online!



Miscellaneous

- Long hold a vowel to open a popup with accented variants.
- ⌥ + any letter makes symbols.
- ⇧ + ⌥ + any letter makes even more symbols.
  - For example, to make “Pretty quotes”: (⇧ +) ⌥ + [
- Application settings standard shortcut is ⌘ + ,
  - If you try ⌘ + P like on Windows, that's the *Print* command instead.
- ⌥ + ⌘ + D: toggle dock auto-hide.

Gestures

Gesture	Action
Three fingers up	Show all windows across all apps (and spaces).
Three fingers down	Show all windows in current app.
Swipe three fingers to the left/right	Switch spaces (or switch to full screen app).
Place four fingers down then spread them	Show desktop ().
Drag two fingers in any direction in a scrollable area	Scroll (direction depends on “natural direction” setting).
Drag two fingers to the left/right	Navigate to previous/next page (in a browser).
Drag two fingers from the right, outside of the trackpad	Show notifications.



IV. Pitfalls

Tiny chapter to make it a nice three-column table of contents.

Common issues

If you can't find your app:

- You might have minimized it with ⌘ + M or the  button.
  Look at your dock in the bottom right area.
- You might have full screened it with ^ + ⌘ + F or the  button.
  Look at your spaces by swiping three fingers around.

You have a lot of apps running for no reason. Tabbing to them shows nothing:

- You closed all the windows of that app using ⌘ + W or the  button.
  Closing windows does not quit apps.
  Quit the app with ⌘ + Q or click the main menu of that app (just right of the Apple logo) and choose “Quit”

Advice

- Don't minimize, and only full-screen for videos or presentations.
- Keep all your windows in one space. To find your windows, do either:
  - A combination of ⌘ + Tab and ⌘ + < / >.
  - A combination of ⌘ + Tab and the three fingers down gesture.
  - Three fingers up gesture (if you have a very recognizable window).
- When you intend to quit an app, don't use the  button. Quit the app instead.

V. Third-party tools and applications

The rest of the batteries.

Preamble

Everything after these slides will be automatically installed during the last chapter.

These slides are for demos, and later for reference.

Homebrew

Description	De facto standard package manager for macOS, maintained by the community.
Model	FOSS
Link	https://brew.sh
Install	`/bin/bash -c "$(curl -fsSL https://raw.githubusercontent.com/Homebrew/install/HEAD/install.sh)"`

Notes:

- Main function: install packages and dependencies
- Secondary function: install GUI applications (called “casks”)

iTerm2

Description	`Terminal.app` sucks. Use this instead.
Model	FOSS
Link	https://iterm2.com
Install	`brew install --cask iterm2`

Good alternatives:

- Ghostty
- kitty
- Alacritty

- Best feature: global hotkey. I set it to ⌘ + § .
- Second best-feature: automatic copying.

Useful shortcuts:

- ⌘ + T: New tab.
- ⌘ + K: Clear.
- ⌘ + D: Split vertical.
- ⌘ + ⇧ + D: Split horizontal.
- ^ + D: Close tab/panel (standard UNIX for logging out).

Sublime Text

Description	Lean, fast and customizable text editor.
Model	Paid, with eternal free trial
Link	https://www.sublimetext.com
Install	`brew install --cask sublime-text`

Good alternatives:

- neovim for vim people.
- Zed

Built-in TextEdit app not recommended.

- Not a replacement to IDEs, just a good text editor.
- Nice as a default for opening text-based files.
- Many plugins and themes.
- Very easy to hack via Python.
- Opens instantly.
- Persists all windows and tabs.
- Very good multi-line editing (invented the concept!).

VI. Third-party extensions

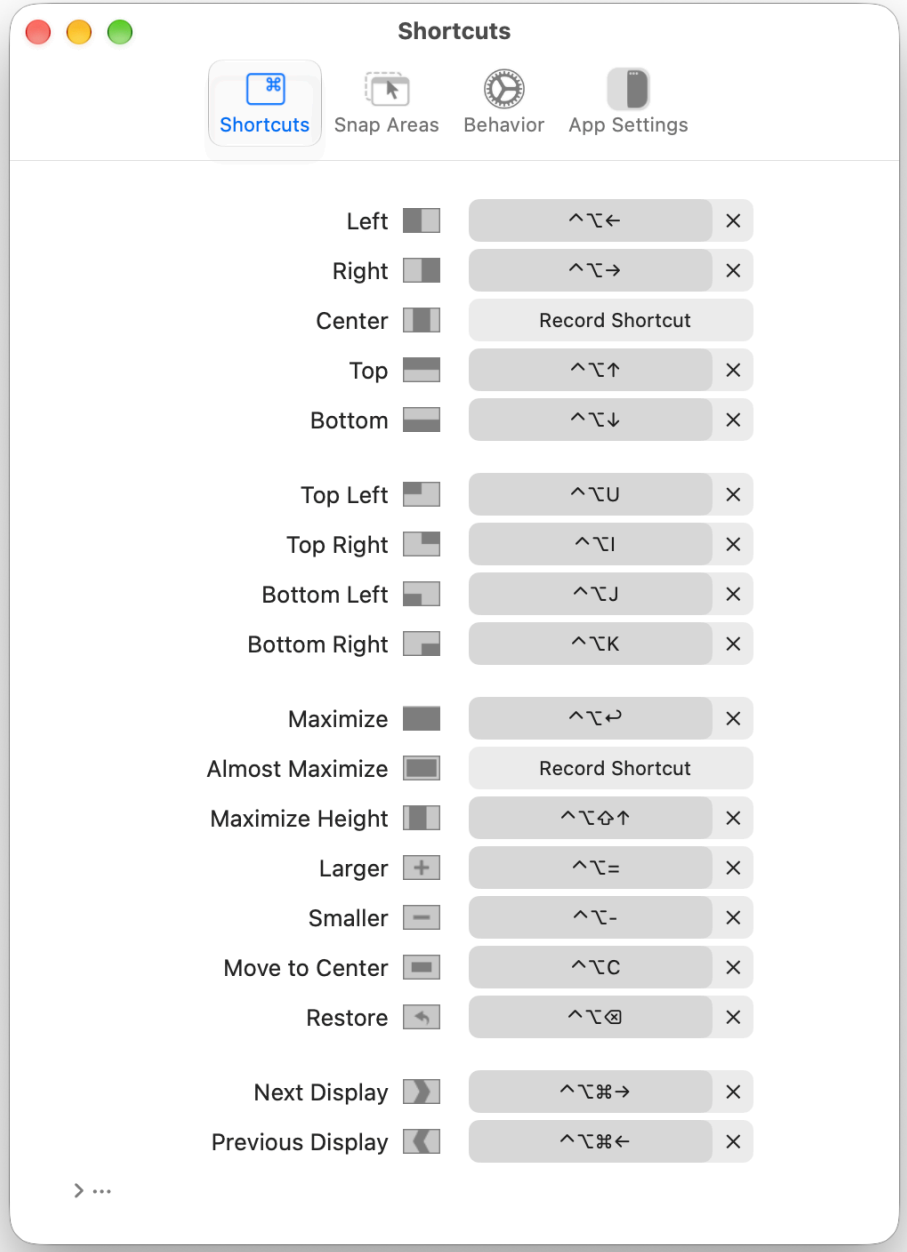
Unofficial tools to customize the OS more, or add missing functionalities.

Rectangle

Description	Window management app
Model	FOSS
Link	https://github.com/rxhanson/Rectangle
Install	`brew install --cask rectangle`

Notes:

- MacOS has its own shortcuts, but the animations are long, and the shortcuts not customizable or natural.
- My favorite shortcut prefix for this app is ⌥ + ⌘.
- I said we'll come back to it later: to maximize, I don't double click a title bar, I do ⌥ + ⌘ + F.

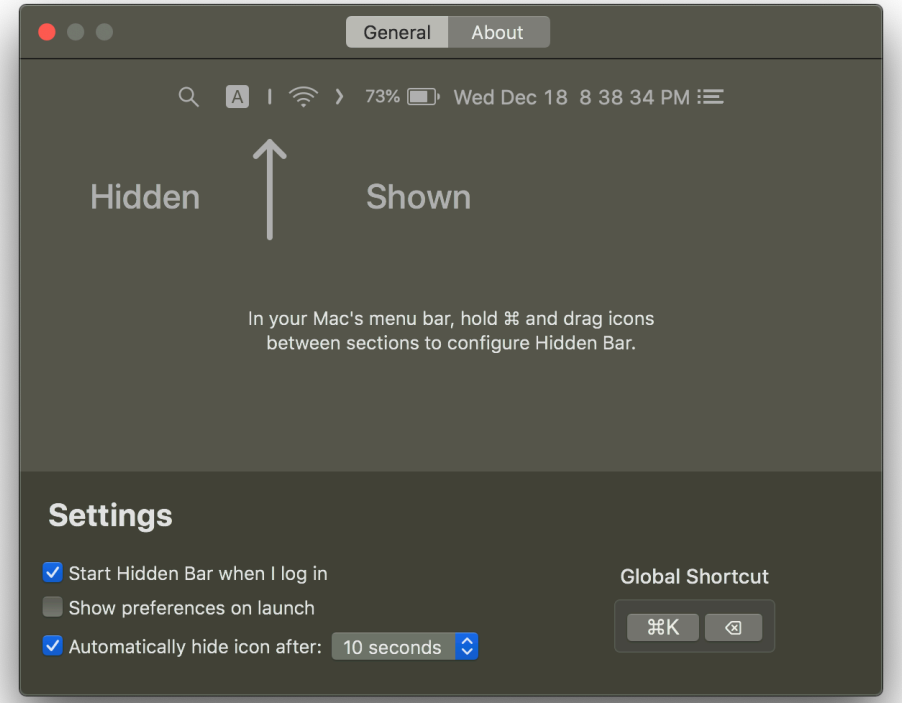


Hidden Bar

Description	Hides your excess menu bar icons
Model	FOSS
Link	https://github.com/dwarvesf/hidden
Install	`brew install --cask hiddenbar`

Notes:

- Can't believe this is not in standard macOS.
- Click the arrow to expand.
- Right click the arrow to show super-hidden icons.

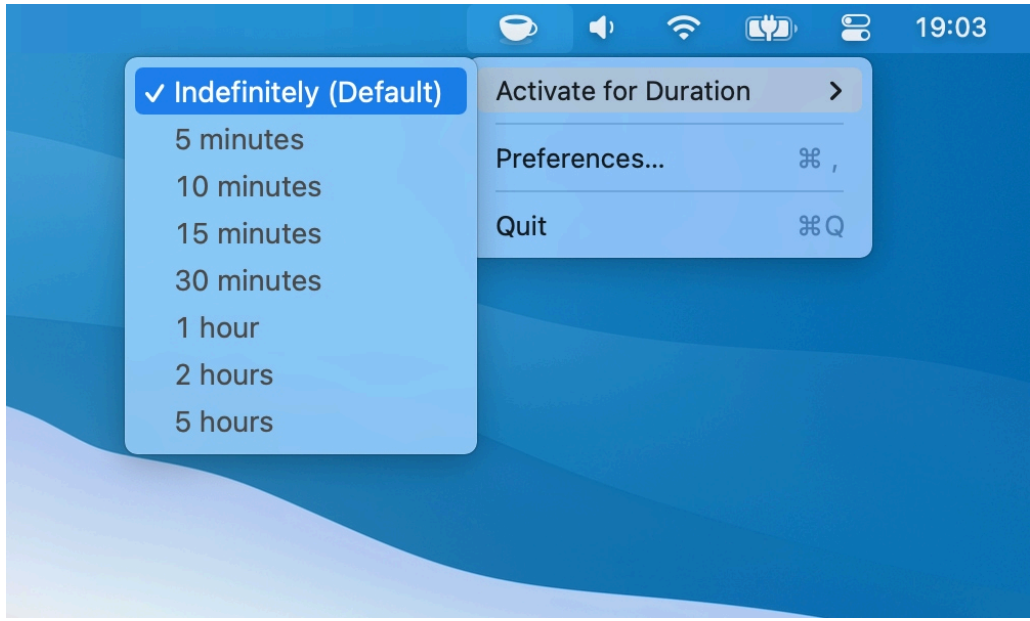


Keeping You Awake

Description	Prevents your Macbook from sleeping
Model	FOSS
Link	https://github.com/newmarcel/KeepingYou
Install	`brew install --cask keepingyouawa`

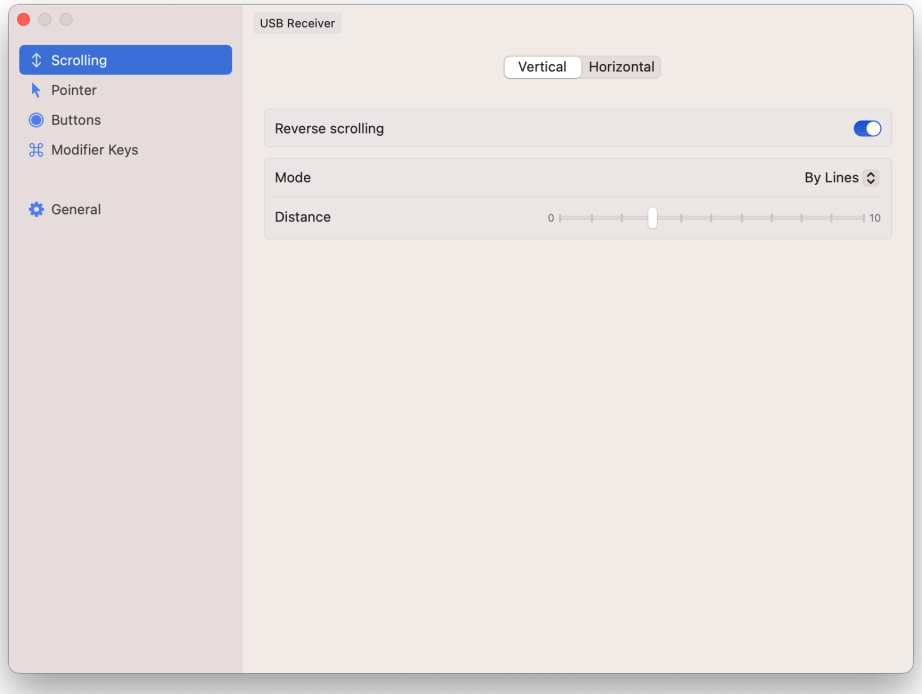
Notes:

- Quick toggle for letting your Mac sleep on idle or not.



Linear Mouse

Description	Disables acceleration on physical scroll wheels
Model	FOSS
Link	https://github.com/linearmouse/linearmous
Install	`brew install --cask linearmouse`

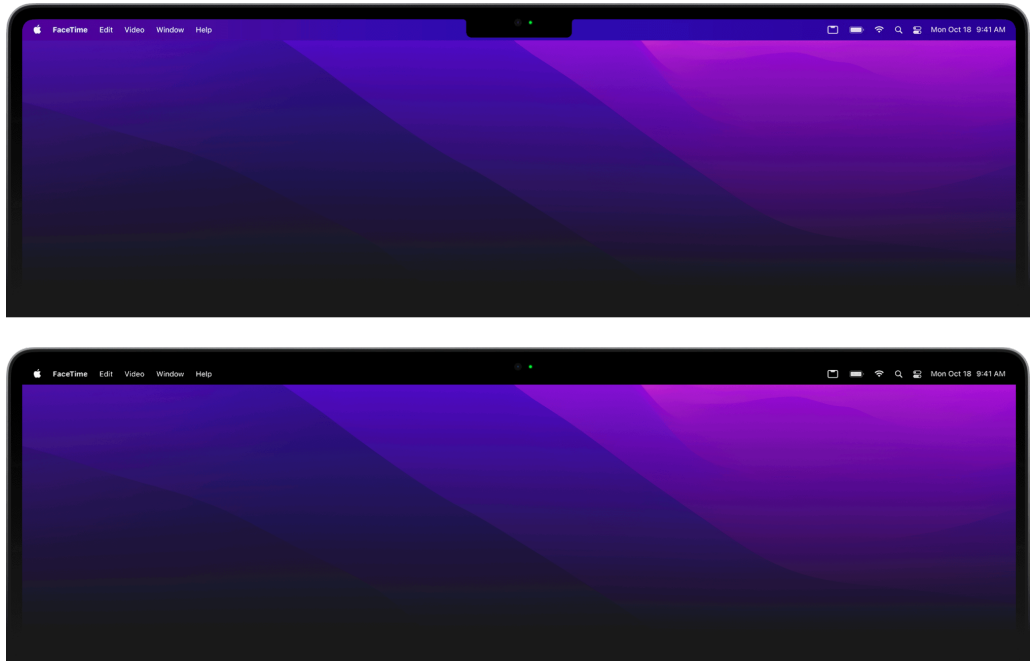


Notes:

- On macOS, the faster you scroll, the more it scrolls.
  If you scroll just one tick, it will literally move one pixel.
- With this app:
  One mousewheel tick = a fixed number of lines.

Top Notch

Description	Hides the stupid notch
Model	Free, but not open source
Link	https://topnotch.app
Install	`brew install --cask topnotch`



Notes:

- To make laptops uslessly sexier, Apple reduced the bezel by adding a thick ugly notch.
- This modifies your wallpaper and adds a black bar.
- Not only you can't see the notch, but it actually looks cooler.
- Sometimes it forgets to do it, just click the menu bar icon.

SwitchResX

Description	Switch any monitor to any resolution
Model	Paid: 14€ lifetime license
Link	https://www.madrau.com
Install	`brew install --cask switchresx`

Notes:

- Apple thinks resolutions are scary for casuals, so they just give DPI scaling options.
- With this you can control exactly what resolution, scaling and refresh rate for anything plugged into your Mac.
- Very useful for presentations.

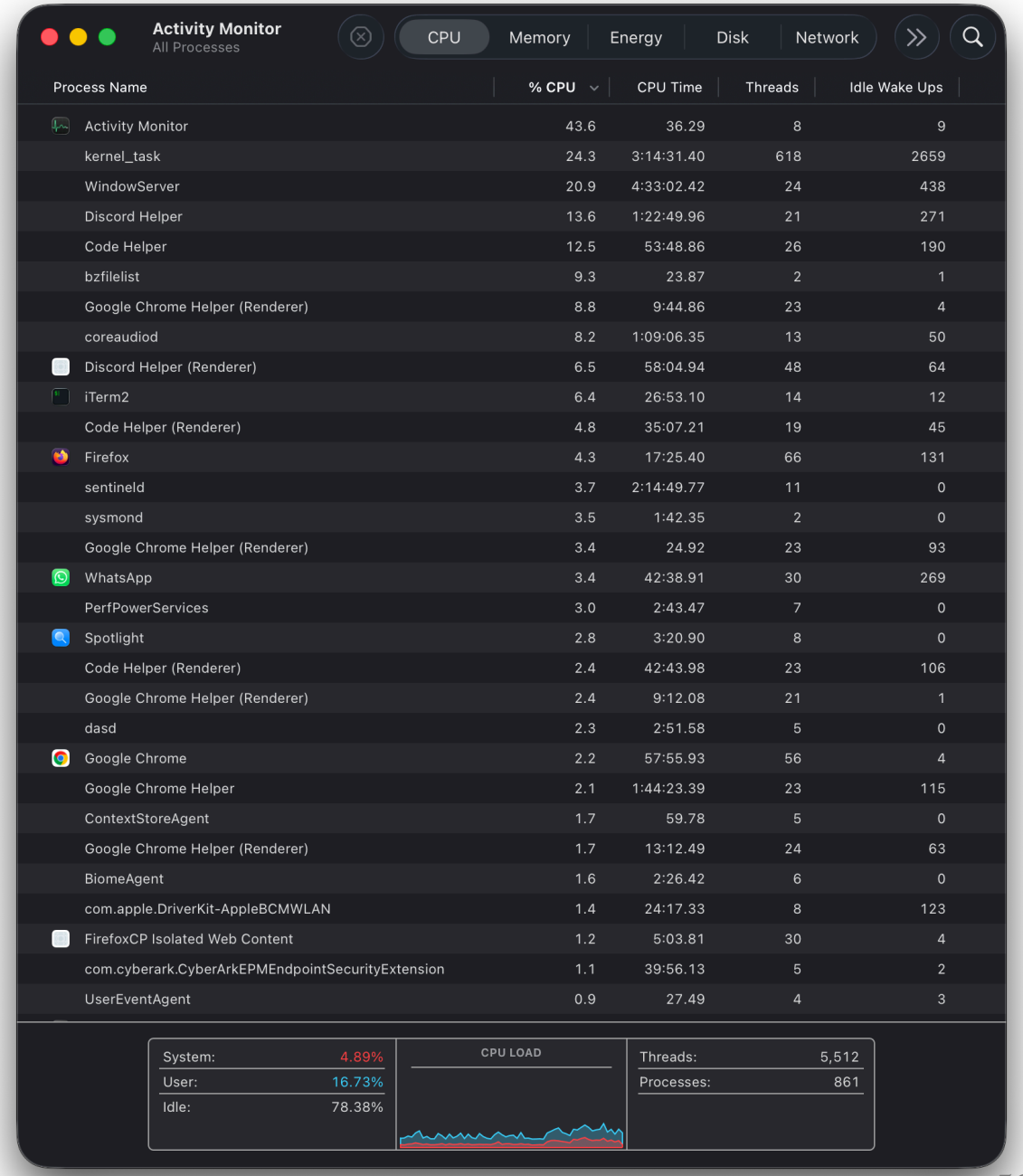


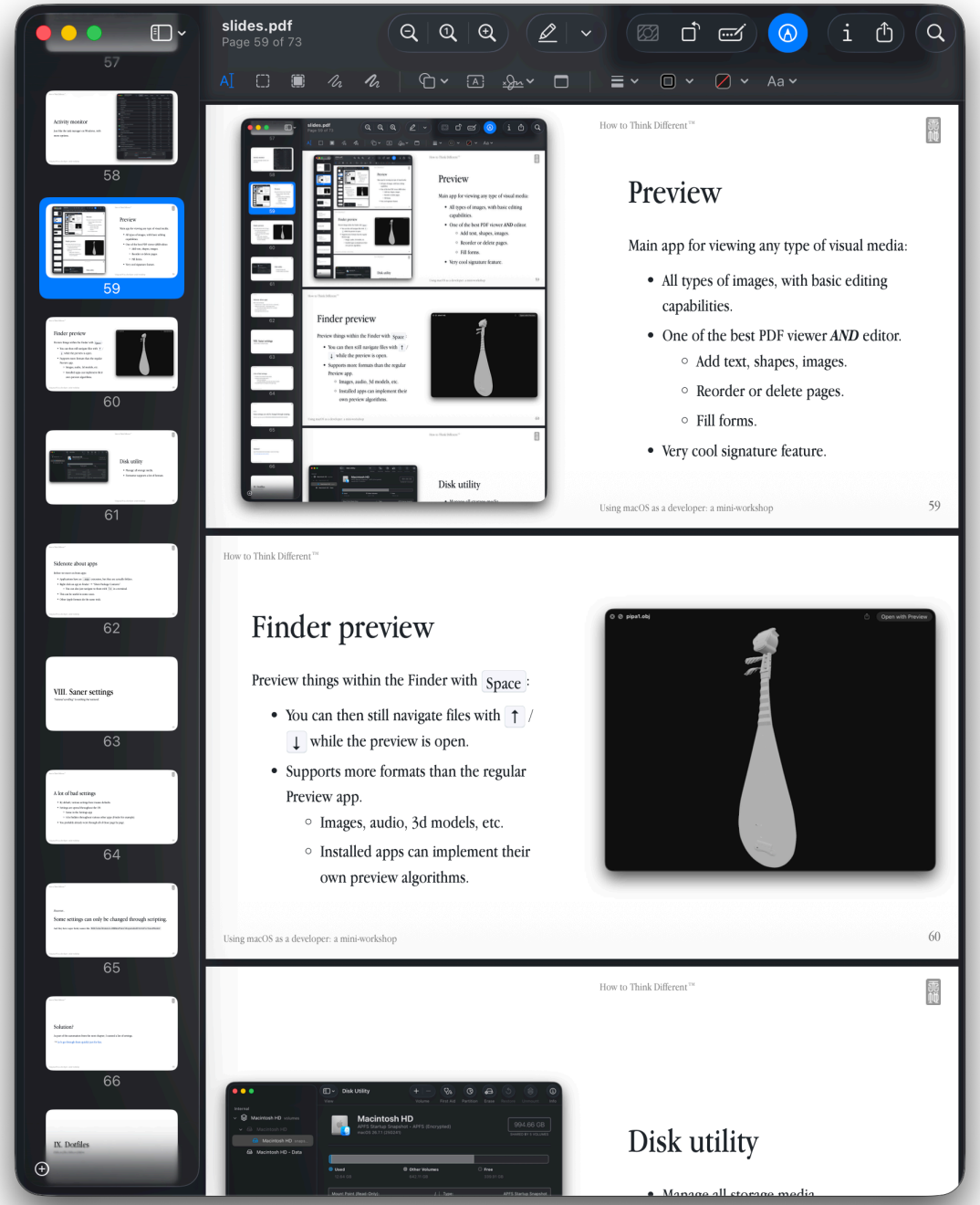
VII. Useful built-in applications

Comes with some batteries included.

Activity monitor

Just like the task manager on Windows, with more options.





Preview

Main app for viewing any type of visual media:

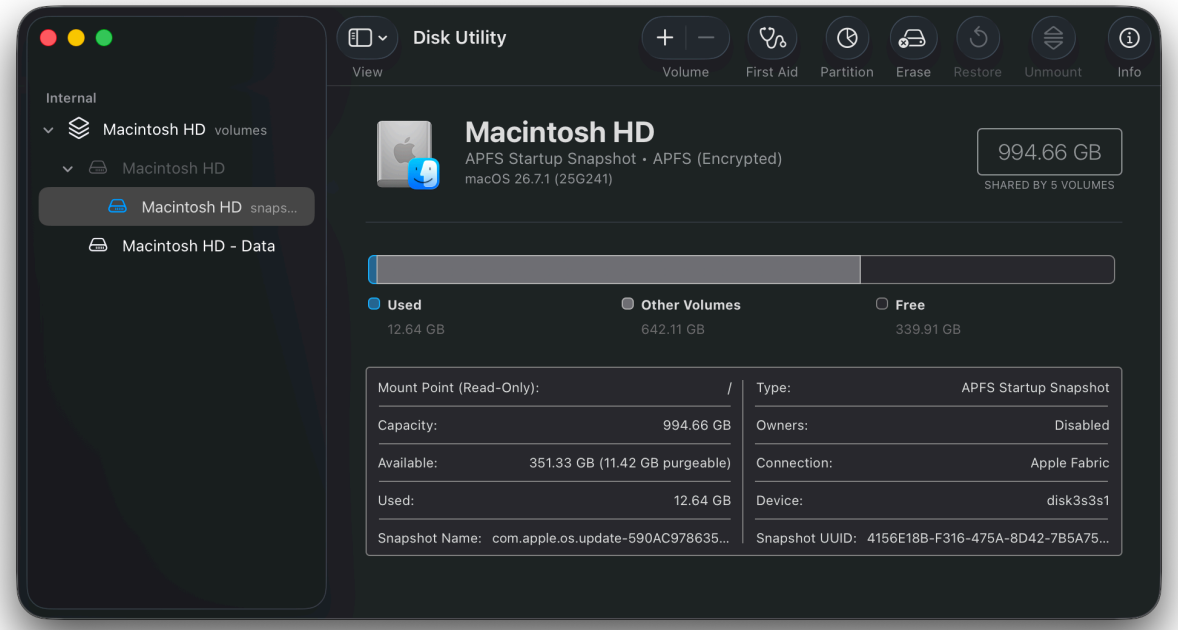
- All types of images, with basic editing capabilities.
- One of the best PDF viewer ***AND*** editor.
  - Add text, shapes, images.
  - Reorder or delete pages.
  - Fill forms.
- Very cool signature feature.

Finder preview

Preview things within the Finder with `Space`:

- You can then still navigate files with `↑` / `↓` while the preview is open.
- Supports more formats than the regular Preview app.
  - Images, audio, 3d models, etc.
  - Installed apps can implement their own preview algorithms.





Disk utility

- Manage all storage media.
- Formatter supports a lot of formats.

Sidenote about apps

Before we move on from apps:

- Applications have an `.app` extension, but they are actually folders.
- Right click an app in Finder → “Show Package Contents”.
    - You can also just navigate to them with `ls` in a terminal.
- This can be useful in some cases.
- Other Apple formats do the same trick.

VIII. Saner settings

“Natural scrolling” is nothing but natural.

A lot of bad settings

- By default, various settings have insane defaults.
- Settings are spread throughout the OS:
  - Some in the Settings app
  - A lot hidden throughout various other apps (Finder for example)
- You probably already went through all of those page by page.

However...

Some settings can only be changed through scripting.

And they have super funky names like `NSGlobalDomain.NSNavPanelExpandedStateForSaveMode2`

Solution?

As part of the automation from the next chapter, I curated a list of settings.

→ Let's go through them quickly just for fun.

IX. Dotfiles

Hide yo files, hide yo folders.

What are dotfiles?

- On UNIX-like environments, hidden files or folders.
- Most tools will pollute your `~/` full of dotfiles.
- Configuring a whole new computer can take a week.
  - People have started version controlling their home folder dotfiles.
  - Even better: they made tools to manage and apply their version-controlled dotfiles.
  - It's like CI for your own computer.

Chezmoi

- Tool for applying dotfiles easily.
- Every file/script is templatable.
- Supports various triggers for executing scripts.
- Support for many password managers.
- Explicit name-based permissions: private files, executables, etc.
- Dotfiles made by Chezmoi are symlinks
  → **Change your settings in an app, the change is made directly in your repository**.
- And many other features I don't use.

My dotfiles

- All my configs, settings, custom scripts, in one repo: https://github.com/Tenchi2xh/dotfiles
- What it does:
  - Installs apps, extensions and CLI tools,
  - Setup custom file associations,
  - Configure macOS with custom settings,
  - Configure `zsh` with custom theme, animations and plugins,
  - Configure some third-party tools,
  - Installs and uses my own customized terminal font,
  - Optional: organize the dock and set up custom icons.
- Very opinionated and themed around my branding, but easily customizable.

Requirements

- Install Homebrew:

```
/bin/bash -c "$(curl -fsSL https://raw.githubusercontent.com/Homebrew/install/HEAD/install.sh)"
```

- Install Chezmoi: `brew install chezmoi`.
- If using Bitwarden: `brew install bitwarden-cli`.
- Create a `SSH Key` entry in bitwarden, add your private SSH key in the notes field.
  → If not desired, delete `dotfiles/private_dot_ssh/private_id_rsa.tmpl`
- Create an `Auth tokens` entry in bitwarden, add a custom field `HOMEBREW_GITHUB_API_TOKEN`.
  → If not desired, delete the line `export HOMEBREW_GITHUB_API_TOKEN=...` in `dotfiles/dotfiles/dot_zprofile.tmpl`.

How to use them

- Fork my repo https://github.com/Tenchi2xh/dotfiles.
- Leave a ★.
- In the `config/` folder, remove or edit what you don't need from the YAMLs (leave empty lists when needed).
- In the `scripts/` folder, remove scripts you don't care about (for example Dock customization, custom icons).
- In the `dotfiles/` folder, remove config files that don't apply to you (also replace the public SSH key).
- `commit` and `push`.
- If using Bitwarden: `bw login` and `bw unlock`.
- Apply: `chezmoi init --apply [YOUR_GITHUB_USERNAME]`



Peace.